

ACG-MAC — Adaptive Communication-aware Graph Model for Action Coordination

Implemented decision-surrogate member · selected hidden width $h = 56$ · final reported inference averages $K = 5$ independently seeded members

01 · Prepared Inputs

Node state tensor
 $X \in \mathbb{R}^{69 \times 10}$
 P,Q · nominal P,Q ·
 degree/depth
 availability · normalized age
 · slack flag · node flag

Electrical graph
 $A \in \{0, 1\}^{69 \times 69}$
 node mask $m_n \in \{0, 1\}^{69}$
 radial candidate topology

System context
 $\tilde{z} \in \mathbb{R}^{21}$
 train-standardized
 global, telemetry,
 physics-proxy and
 topology descriptors

Frozen physical anchors
 $V_{\min}^{proxy}, L^{proxy}$
 Linear DistFlow
 approximation
 non-trainable reference
 path

02 · Telemetry-Quality-Aware Graph Encoder

[Trainable] Node projection
 $\mathbb{R}^{10} \rightarrow \mathbb{R}^{56}$
 $H_0 = \text{ReLU}(W_{in}X + b)$

[Deterministic] Graph normalization
 $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$
 masked padded nodes
 excluded

[Trainable] Two graph-propagation steps
 $H_1 = \text{ReLU}(\hat{A}H_0)$
 $H_2 = \text{ReLU}(W_hH_1 + b_h)$
 $H_3 = \text{ReLU}(\hat{A}H_2)$

[Trainable] Learned telemetry weighting
 $[o_i, \tilde{a}_i] \in \mathbb{R}^2 \rightarrow 12 \rightarrow 1$
 $\tilde{a}_i = a_i/4, \quad r_i = \sigma(W_2 \text{ReLU}(W_1[o_i, \tilde{a}_i]))$

Telemetry-quality-weighted pooling
 $\alpha_i = \frac{r_i m_i}{\sum_j r_j m_j}$
 $\bar{h} = \sum_i \alpha_i H_{3,i} \in \mathbb{R}^{56}$
 $h_{\max} = \max_i H_{3,i} \in \mathbb{R}^{56}$

03 · Graph-Context Fusion Core

Graph summary
 $\bar{h}, h_{\max} \in \mathbb{R}^{56}$

System context
 $\tilde{z} \in \mathbb{R}^{21}$

Context concatenation
 $c = [\bar{h}; h_{\max}; \tilde{z}] \in \mathbb{R}^{133}$
 $56 + 56 + 21 = 133$

[Trainable] Fusion MLP
 $133 \rightarrow 96 \rightarrow 48$
 ReLU · Dropout = 0.08 · ReLU
 latent decision state $u \in \mathbb{R}^{48}$

Selected full configuration
direct fusion is active (simple_fusion = True)
adaptivity is supplied by telemetry-quality-weighted graph pooling

04 · Physics-Residual and Feasibility Heads

[Trainable] Residual regression head
 Linear : $48 \rightarrow 2$
 $[\Delta V, \Delta \ell] = W_r u + b_r$
 $\Delta \ell$ acts in $\log(1 + L)$ domain

[Frozen] Physics reconstruction
 $\hat{V}_{\min} = V_{\min}^{proxy} + \Delta V$
 $\hat{L} = \max(0, \exp[\log(1 + L^{proxy}) + \Delta \ell] - 1)$

[Trainable] Feasibility head
 Linear : $48 \rightarrow 1$
 $p_f = \sigma(w_f^T u + b_f)$

ACG-MAC member output
 $\hat{y} = [\hat{V}_{\min}, \hat{L}, p_f]$
 decision-support surrogate output used to rank candidate actions

Training / Final Inference Modes

TRAINING ONLY
 $\mathcal{L}_{reg} = \text{mean}[\frac{((\hat{y}_{res} - y_{res}))/[0.05, 1]}{[0.05, 1]}]^2$
 $\mathcal{L} = \mathcal{L}_{reg} + 0.18 \text{BCE}(c)$
 AdamW · lr = 1.8×10^{-3} · weight decay = 2×10^{-4} · early-stopping patience = 6

FINAL REPORTED INFERENCE
 $\bar{y} = \frac{1}{K} \sum_{k=1}^K \hat{y}^{(k)}, \quad K = 5 \text{ independently seeded members}$